

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI – 590 018



**An Internship Project Report
on**

Online Marketplace

Submitted in partial fulfillment of the requirements for the VIII Semester of
degree of **Bachelor of Engineering in Information Science and Engineering** of
Visvesvaraya Technological University, Belagavi

by

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Under the Guidance of

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ESTD:2001
An Institute with a Difference

Department of Information Science and Engineering

RNS Institute of Technology

**Dr. Vishnuvaradhan Road, Rajarajeshwari Nagar post,
Channasandra, Bengaluru-560098**

2022-2023

RNS INSTITUTE OF TECHNOLOGY

Dr. Vishnuvaradhan Road, Rajarajeshwari Nagar post,
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DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING



CERTIFICATE

Certified that the Internship work entitled **Online Marketplace** has been successfully completed by **Amardeep Kumar (1RN19IS020)** and **Devendra Upadhyay (1RN19IS056)** a bonafide student of **RNS Institute of Technology, Bengaluru** in partial fulfillment of the requirements of 7th semester for the award of degree in **Bachelor of Engineering in Information Science and Engineering of Visvesvaraya Technological University, Belagavi** during academic year **2022-2023**. The internship report has been approved as it satisfies the academic requirements in respect of internship work for the said degree.

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Name of the Examiners

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Signature with Date

1. _____

1. _____

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DECLARATION

We, **Amardeep Kumar [USN: 1RN19IS020]** and **Devendra Upadhyay[USN: 1RN19IS056]** students of VII Semester BE, in Information Science and Engineering, RNS Institute of Technology hereby declare that the Internship work entitled ***Online Marketplace*** has been carried out by us and submitted in partial fulfillment of the requirements for the *VII Semester degree of **Bachelor of Engineering in Information Science and Engineering** of Visvesvaraya Technological University, Belagavi* during academic year 2022-2023.

Place : Bengaluru

Date :

Amardeep Kumar
(1RN19IS020)

Devendra Upadhyay
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Date: September 16th 2022

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **Mr. Amardeep Kumar** a student from RNSIT with USN: 1RN19IS020 has successfully completed Internship program in **Full Stack Development** from August 2022 till September 2022.

During this period of internship, Mr. Amardeep Kumar worked on various areas of software development including project work titled "**Online Marketplace**". He has successfully met the objectives of the internship program and his conduct was found to be satisfactory.

We wish him all the best for future endeavors.



Kiran Gopalakrishna,
Founder & CEO,
TechieAid, Bangalore

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During this period of internship, Mr. Devendra Upadhyay worked on various areas of software development including project work titled "**Online Marketplace**". He has successfully met the objectives of the internship program and his conduct was found to be satisfactory.

We wish him all the best for future endeavors.



Kiran Gopalakrishna,
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ABSTRACT

Internet Advertising is the way of the future and it is very evident since many companies and businesses have their own web sites and advertisements are located all over the World Wide Web. The Internet or World Wide Web is quickly becoming the most effective way for a business to advertise their products or services to customers.

Online Marketplace is classified ads website. Here, user can create their account and post ads with images. Other user can look at these ads and contact the seller. The seller can update as well as delete their ads. The admin can delete the users ads if found inappropriate or fake. This website is made using technology such as html, css, js, ejs, node and mongodb.

ACKNOWLEDGMENT

The fulfillment and rapture that go with the fruitful finishing of any assignment would be inadequate without the specifying the people who made it conceivable, whose steady direction and support delegated the endeavors with success.

We would like to profoundly thank **Management of RNS Institute of Technology** for providing such a healthy environment to carry out this Internship work with project.

We would like to express our thanks to our Principal **Dr. M K Venkatesha** for his support and inspired us towards the attainment of knowledge.

We wish to place on record our words of gratitude to **Dr. Suresh L**, Professor and Head of the Department, Information Science and Engineering, for being the enzyme and master mind behind our Internship work with project.

We would like to express our heartfelt thanks to our Internship Guide, **Ms. Shyla N**, Assistant professor, Department of Information Science and Engineering for always helping and being there whenever we needed her guidance.

We would like to express our profound and cordial gratitude to our Internship Coordinators, **DR. R Rajkumar**, Associate Professor, and **Mr. Pramoda R**, Assistant Professor, Department of Information Science and Engineering for their valuable guidance, constructive comments, continuous encouragement throughout the Internship Work and guidance in preparing report.

We would like to thank all other teaching and non-teaching staff of Information Science & Engineering who have directly or indirectly helped us to carry out the Internship Work.

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CHAPTER 1

INTRODUCTION

1.1 ORGANIZATION

1.1.1 Company Profile

TechieAid is a high-end software innovation company based in Bangalore, India focused on Full Stack Web Applications and Mobile App Development that partners with self-funded start-ups to Fortune 500 companies. With over a decade's experience in technology, design, and product development, we're experts at delivering digital solutions using cutting-edge technology that solves real-time problems for our clients. Over the last 4 years, TechieAid's passionate team has partnered with our clients to make immersive digital solutions that completely transform their businesses. We are more than just consultants, we put our clients at the heart of what we do to add real value to their business.

1.1.2 Domain

Server-Side Development using NodeJS and MongoDB

Server-side scripting is a technique used in web development which involves employing scripts on a web server which produce response customized for each user's (client's) request to the website. These scripts may assemble client characteristics for use in customizing the response based on those characteristics, the user's requirements, access rights, etc. Server-side scripting also enables the website owner to hide the source code that generates the interface, whereas with client-side scripting, the user has access to all the code received by the client.

The development stack used:

- MongoDB
- NodeJS
- EJS

1.2 PROBLEM STATEMENT

“This project is aimed to automate the online marketplace. The purpose of the project entitled as ONLINE MARKETPLACE is to computerize the classified ads system to develop software, which is user friendly, simple, fast, and cost effective. The main function of the system is to register and store user details and also their advertisement post.”

1.2.1 Existing System and their Limitations

The existing system is not user friendly because a lot of privacy breach and fake ads exist. Also, the data is not stored efficiently. A lot of times there are lots of redundant data in the system which often makes it slow and inaccurate.

Limitations:

- **Inefficient storage and backup method:** Use of non cloud based method to store the data.
- **Privacy breach:** the user details are stored inefficiently leading to hacks or the data are sold to third party organisation.
- **Lots of fake ads:** in todays system a lot of fake ads are present where seller scams the buyer.

1.2.2 Proposed Solution

The proposed solution is a web application called “Online Marketplace” which helps the users to overcome the problems faced in the existing system. The Online Marketplace uses authentication system via stored email and password which are used to authenticate for users to log into the system. The system is accessible by admin, seller and buyer. The admin can easily monitor the fake ads of the seller and delete them.

The project online marketplace uses the hashing encryption method to store the data of the user. Using this way, we can easily maintain the privacy of the user and keep them safe from any type of privacy breach. Even the admin cannot see the user details such as password. Thus they cannot leak any data to any third party organisation.

This project makes use of the cloud service-as-a-system to store all the images of the users and thus can easily maintained and manage data. It also helps in fast retrieval, storage capacity management and easily monitoring the redundant data.

CHAPTER 2

REQUIREMENT ANALYSIS, TOOLS & TECHNOLOGIES

2.1 HARDWARE AND SOFTWARE REQUIREMENTS

2.1.1 Hardware requirements

Processor	:	Pentium4 processor
Processor Speed	:	2.4 GHz
RAM	:	1 GB
Storage Space	:	10 GB

2.1.2 Software Requirements

Operating System: Windows

Browser: Mozilla Firefox, Google Chrome, Microsoft Edge, Opera

2.2 TOOLS and TECHNOLOGIES

2.2.1 INTRODUCTION TO WEB

Internet is a global communication system that links together thousands of individual networks. It allows exchange of information between two or more computers on a network. Thus, internet helps in transfer of messages through mail, chat, video & audio conference, etc. It has become mandatory for day-to-day activities: bills payment, online shopping and surfing, tutoring, working, communicating with peers, etc.

The World Wide Web (WWW), commonly known as the Web, is an information system where documents and other web resources are identified by Uniform Resource Locators (URLs, such as <https://example.com/>), which may be interlinked by hypertext, and are accessible over the Internet. The resources of the Web are transferred via the Hypertext Transfer Protocol (HTTP), may be accessed by users by a software application called a web browser, and are published by a software application called a web server. The World Wide Web is not synonymous with the Internet, which pre-existed the Web in some form by over two decades and upon which technologies the Web is built.

Web consists of billions of clients and server connected through wires and wireless networks. The web clients make requests to web server. The web server receives the request finds the resources

and return the response to the client. When a server answers a request, it usually sends some type of content to the client. The client uses web browser to send request to the server. The server often sends response to the browser with a set of instructions written in HTML (Hyper Text Markup Language). All browsers know how to display HTML page to the client.

Web resources may be any type of downloaded media, but web pages are hypertext documents formatted in Hypertext Markup Language (HTML). Special HTML syntax displays embedded hyperlinks with URLs which permits users to navigate to other web resources. In addition to text, web pages may contain references to images, video, audio, and software components which are either displayed or internally executed in the user's web browser to render pages or streams of multimedia content.

Multiple web resources with a common theme and usually a common domain name, make up a website. Websites are stored in computers that are running a web server, which is a program that responds to requests made over the Internet from web browsers running on a user's computer. Website content can be provided by a publisher, or interactively from user-generated content. Websites are provided for a myriad of informative, entertainment, commercial, and governmental reasons.

2.2.2 HTML

HTML (Hypertext Markup Language) is not a programming language. It is a markup language that tells web browsers how to structure the web pages you visit. It can be as complicated or as simple as the web developer wants it to be. HTML consists of a series of elements, which you use to enclose, wrap, or markup different parts of content to make it appear or act in a certain way. The enclosing tags can make content into a hyperlink to connect to another page, italicize words, and so on. It is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document.

HTML can embed programs written in a scripting language such as JavaScript, which affects the behavior and content of web pages. Inclusion of CSS defines the look and layout of content. The World Wide Web Consortium (W3C), former maintainer of the HTML and current maintainer of the CSS standards, has encouraged the use of CSS over explicit presentational HTML since 1997.

HTML elements are the building blocks of HTML pages. With HTML constructs, images

and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as `<img />` and `<input />` directly introduce content into the page. Other tags such as `<p>` surround and provide information about document text and may include other tags as sub - elements. Browsers do not display the HTML tags but use them to interpret the content of the page.

HTML markup consists of several key components, including those called tags (and their attributes), character-based data types, character references and entity references. HTML tags most come in pairs like `<h1>` and `</h1>`, although some represent empty elements and so are unpaired, for example `<img>`. The first tag in such a pair is the start tag, and the second is the end tag (they are also called opening tags and closing tags).

2.2.3 JavaScript

JavaScript, often abbreviated as JS, is a light weighted or just in time compiled programming language that conforms to the ECMAScript specification with first class functions. JavaScript is high-level, often just-in-time compiled, and multi- paradigm. It has curly-bracket syntax, dynamic typing, prototype-based object-orientation, and first-class functions. Alongside HTML and CSS, JavaScript is one of the core technologies of the World Wide Web. JavaScript enables interactive web pages and is an essential part of web applications. Most websites use it for client-side page behavior, and all major web browsers have a dedicated JavaScript engine to execute it. Today, JavaScript can execute not only in the browser, but also on the server, or on any device that has a special program called the JavaScript engine. JavaScript's capabilities greatly depend on the environment it's running.

As a multi-paradigm language, JavaScript supports event-driven, functional, and imperative programming styles. It has application programming interfaces (APIs) for working with text, dates, regular expressions, standard data structures, and the Document Object Model (DOM). However, the language itself does not include any input/output (I/O), such as networking, storage, or graphics facilities, as the host environment (usually a web browser) provides those APIs. JavaScript engines were originally used only in web browsers, but they are now embedded in some servers, usually via Node.js. They are also embedded in a variety of applications created with frameworks such as Electron and Cordova. JavaScript is primarily used in the browser, enabling developers to manipulate webpage content through the DOM, manipulate data with AJAX and Indexed DB, draw graphics with canvas, interact with the device running the browser through various APIs, and more.

2.2.4 EJS

EJS or Embedded Javascript Templating is a templating engine used by Node.js. Template engine helps to create an HTML template with minimal code. Also, it can inject data into HTML template at the client side and produce the final HTML. EJS is a simple templating language which is used to generate HTML markup with plain JavaScript. It also helps to embed JavaScript to HTML pages.

EJS Features

- EJS was so similar to HTML so I found it easy.
- EJS uses all the JS jargon and logic, so if you are proficient in JS, you can use EJS right away.
- EJS is way faster than Jade and handlebars.
- EJS has a really smart error handling mechanism built right into it. It points out to you, the line numbers on which an error has occurred so that you don't end up looking through the whole template file wasting your time in searching for bugs.
- Simple template tags: `<% %>`.
- Custom delimiters (e.g., use `<? ?>` instead of `<% %>`)

2.2.5 NodeJS

Node.js is an open-source server environment. Node.js is cross-platform and runs on Windows, Linux, Unix, and macOS. Node.js is a back-end JavaScript runtime environment. Node.js runs on the V8 JavaScript Engine and executes JavaScript code outside a web browser.

Node.js lets developers use JavaScript to write command line tools and for server-side scripting. The functionality of running scripts server-side produces dynamic web page content before the page is sent to the user's web browser. Consequently, Node.js represents a "JavaScript everywhere" paradigm, unifying web-application development around a single programming language, rather than different languages for server-side and client-side scripts.

Advantages of NodeJS

1. It is based on javascript so we can share the code between client side and server side.
2. There will be less conflicts in case of code replacement we can easily replace code file with other code having a better logic without worrying about versioning conflicts.
3. Awesome package management system npm.
4. Requires less memory and utilizes cpu to the maximum extent.

2.2.6 MongoDB

MongoDB is a source-available cross-platform document-oriented database program. Classified as a NoSQL database program, MongoDB uses JSON-like documents with optional schemas. MongoDB is developed by MongoDB Inc. and licensed under the Server Side Public License (SSPL) which is deemed non-free by several distributions.

MongoDB's document model allows virtually any data structure to be modeled and manipulated easily. MongoDB's BSON data format, inspired by JSON, allows you to have objects in one collection with different sets of fields (say, a middle name on a user only when applicable, or region-specific information that only applies to some records).

MongoDB supports creating explicit schemas and validating data. This flexibility is an incredible asset when handling real-world data and changes in requirements or environment.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

3.1.1 MVC Architecture

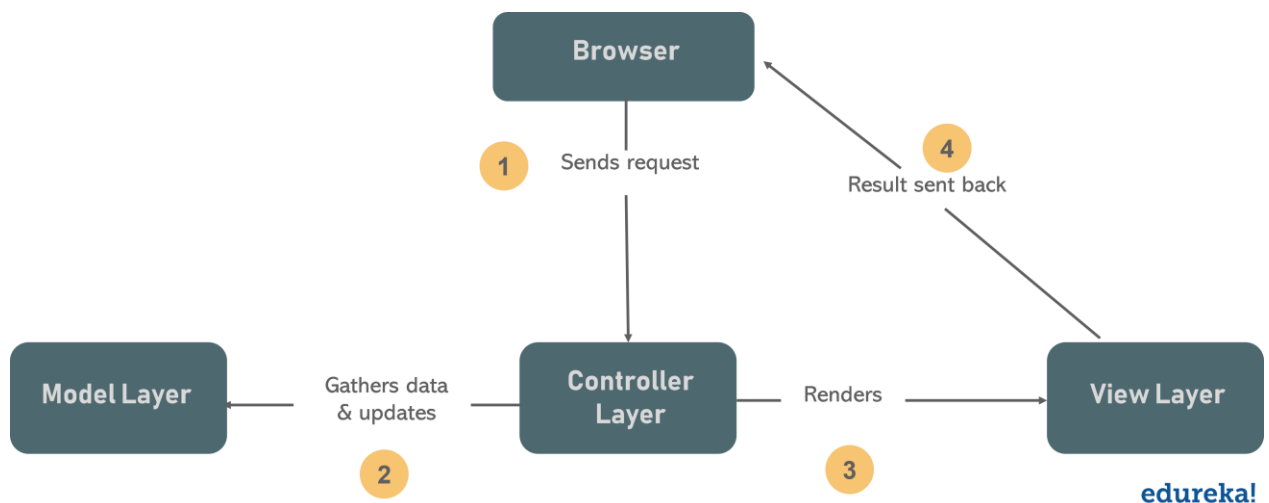


Figure 3.1 MVC architecture

Model

The Model component corresponds to all the data-related logic that the user works with. This can represent either the data that is being transferred between the View and Controller components or any other business logic-related data. For example, a Customer object will retrieve the customer information from the database, manipulate it and update it data back to the database or use it to render data.

View

The View component is used for all the UI logic of the application. For example, the Customer view will include all the UI components such as text boxes, dropdowns, etc. that the final user interacts with.

Controller

Controllers act as an interface between Model and View components to process all the business logic and incoming requests, manipulate data using the Model component and interact with the Views to render the final output. For example, the Customer controller will handle all the

interactions and inputs from the Customer View and update the database using the Customer Model. The same controller will be used to view the Customer data.

3.1.2 Three-tier Architecture

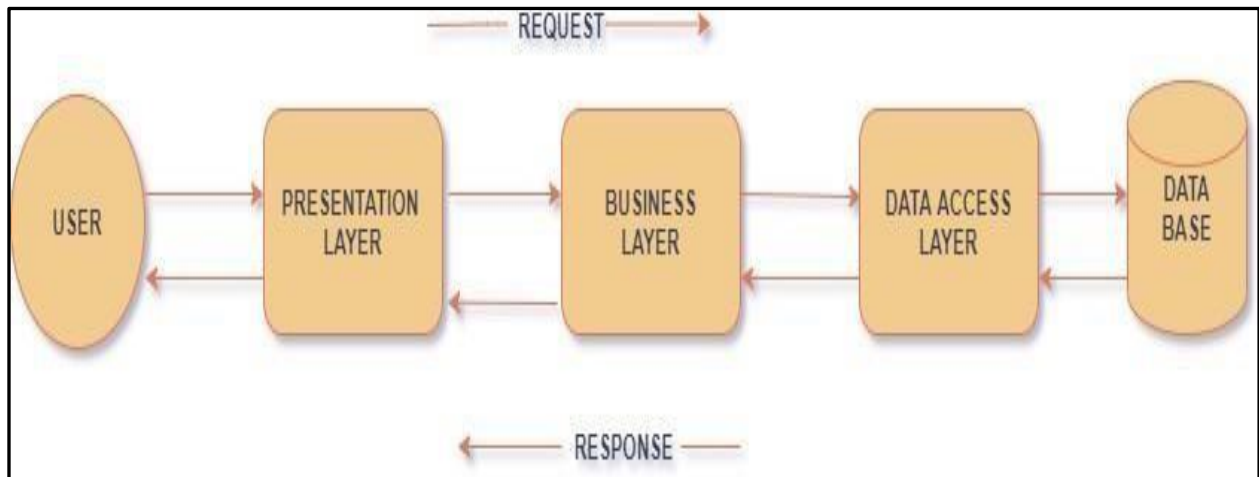


Figure 3.2 Three-tier Architecture

Three-tier architecture is a well-established software application architecture that organizes applications into three logical and physical computing tiers: the presentation tier, or user interface; the application tier, where data is processed; and the data tier, where the data associated with the application is stored and managed.

The chief benefit of three-tier architecture is that because each tier runs on its own infrastructure, each tier can be developed simultaneously by a separate development team, and can be updated or scaled as needed without impacting the other tiers.

Presentation layer: The presentation layer is the user interface and communication layer of the application, where the end user interacts with the application. Its main purpose is to display information to and collect information from the user.

Business layer: The business layer will accept user requests from the browser, processes them, and determines the routes through which the data will be accessed. The workflows by which the data and requests travel through the back end are encoded in a business layer.

Data access layer: This data access layer is a centralized location that receives all data calls and provides access to the persistent storage of an application. This layer is closely connected to the business layer, so the logic knows which database to talk to and the data retrieving process is more optimized.

3.2 Dataflow diagram

User Dataflow

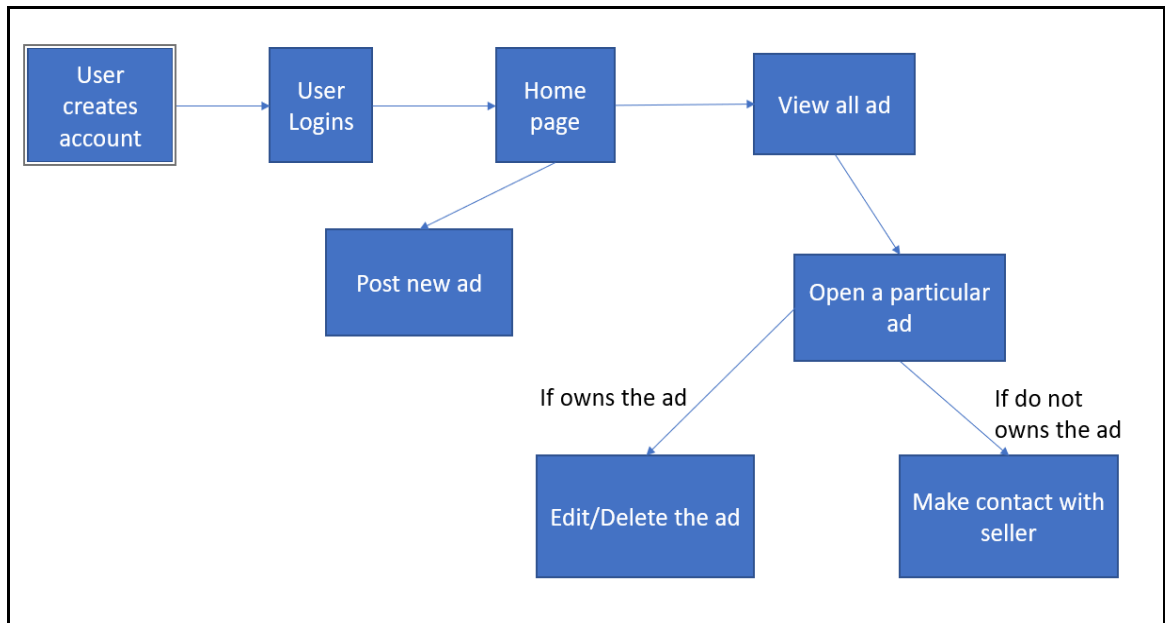


Figure 3.3 User DataFlow Diagram

The user can create a new account by providing details such as email, username and password. Using the username and password provided during new account registration, the user can login. On logging in, home page is shown to the user which provides user with different options such as view all advertisement post, create a new ad post and logout. On clicking the view all ad, the user can see all the ad present in the database. The user can open any particular ad and see its details such as seller description. The user can his own ad by clicking on post new ad option. On filling the form details and uploading it successfully he can see his own ad with the rest of ad in the view all ad option. The user can edit and delete the ad as per his choice.

The admin can login and monitor all the ads. If any ad is found not appropriate he can remove the ad. The admin cannot delete the other users ad.

CHAPTER 4

IMPLEMENTATION

4.1 Functional Modules

4.1.1 HTTP

Hypertext Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML. It was designed for communication between web browsers and web servers, but it can also be used for other purposes. HTTP follows a classical client-server model, with a client opening a connection to make a request, then waiting until it receives a response. HTTP is a stateless protocol, meaning that the server does not keep any data (state) between two requests. Though often based on a TCP/IP layer, it can be used on any reliable transport layer, that is, a protocol that doesn't lose messages silently like UDP does. The Hypertext Transfer Protocol (HTTP) is an application layer protocol for distributed, collaborative, hypermedia information systems.

HTTP functions as a request–response protocol in the client–server computing model. A web browser, for example, may be the client and an application running on a computer hosting a website may be the server. The client submits an HTTP request message to the server. The server, which provides resources such as HTML files and other content, or performs other functions on behalf of the client, returns a response message to the client. The response contains completion status information about the request and may also contain requested content in its message body.

HTTP Methods

HTTP Request	Database Operation	Description
GET	READ	This request is used to get a resource from a server.
POST	CREATE	This request is used to create a new resource on a server.
PUT	UPDATE	This request is used to update a resource on a server.
DELETE	DELETE	This request is used to delete a resource from a server.

4.1.2 REPRESENTATIONAL STATE TRANSFER(REST)

Representational state transfer (REST) is a software architectural style that defines a set of constraints to be used for creating Web services. Web services that conform to the REST architectural style, called RESTful Web services, provide interoperability between computer systems on the internet. RESTful Web services allow the requesting systems to access and manipulate textual representations of Web resources by using a uniform and predefined set of stateless operations.

The Representational State Transfer (REST) style is an abstraction of the architectural elements within a distributed hypermedia system. REST ignores the details of component implementation and protocol syntax in order to focus on the roles of components, the constraints upon their interaction with other components, and their interpretation of significant data elements. It encompasses the fundamental constraints upon components, connectors, and data that define the basis of the Web architecture, and thus the essence of its behavior as a network-based application.

The term REST was introduced by computer scientist Roy Fielding in a dissertation in 2000. Unlike SOAP, which is a protocol, REST is a software architectural style with six constraints for building applications that work over HTTP, often web services. The World Wide Web is the most common realization and application of this architecture style.

4.2 Code Implementation

4.2.1 Database connection

```
const MongoDBStore = require("connect-mongo")(session);
const dbUrl = process.env.DB_URL || "mongodb://localhost:27017/classified";
mongoose.connect(dbUrl, {
  useNewUrlParser: true,
  useCreateIndex: true,
  useUnifiedTopology: true,
  useFindAndModify: false,
});
const db = mongoose.connection;
db.on("error", console.error.bind(console, "connection error:"));
db.once("open", () => {
  console.log("Database connected");
});
```

```
});
```

4.2.2 Ads model

```
const mongoose = require("mongoose");
const Schema = mongoose.Schema;
const ImageSchema = new Schema({
  url: String,
  filename: String,
});
ImageSchema.virtual("thumbnail").get(function () {
  return this.url.replace("/upload", "/upload/w_200");
});
const opts = { toJSON: { virtuals: true } };
const AdsSchema = new Schema(
  {
    title: String,
    images: [ImageSchema],
    price: Number,
    description: String,
    location: String,
    author: {
      type: Schema.Types.ObjectId,
      ref: "User",
    },
    reviews: [
      {
        type: Schema.Types.ObjectId,
        ref: "Review",
      },
    ],
  },
  opts
);
```

4.2.3 User Model

```
const { number } = require('joi');
const mongoose = require('mongoose');
const Schema = mongoose.Schema;
const passportLocalMongoose = require('passport-local-mongoose');

const UserSchema = new Schema({
  admin: Number,
  email: {
    type: String,
    required: true,
    unique: true
  },
});

UserSchema.plugin(passportLocalMongoose);

module.exports = mongoose.model('User', UserSchema);
```

4.2.4 User Controller

```
const User = require("../models/user");
module.exports.renderRegister = (req, res) => {
  res.render("users/register");
};
module.exports.register = async (req, res, next) => {
  try {
    const { email, username, password } = req.body;
    const user = new User({ email, username });
    const registeredUser = await User.register(user, password);
    req.login(registeredUser, (err) => {
      if (err) return next(err);
      req.flash("success", "Welcome to Adify!");
      res.redirect("/ads");
    });
  } catch (e) {
    req.flash("error", e.message);
  }
};
```

```
        res.redirect("register");
    }
};

module.exports.renderLogin = (req, res) => {
    res.render("users/login");
};

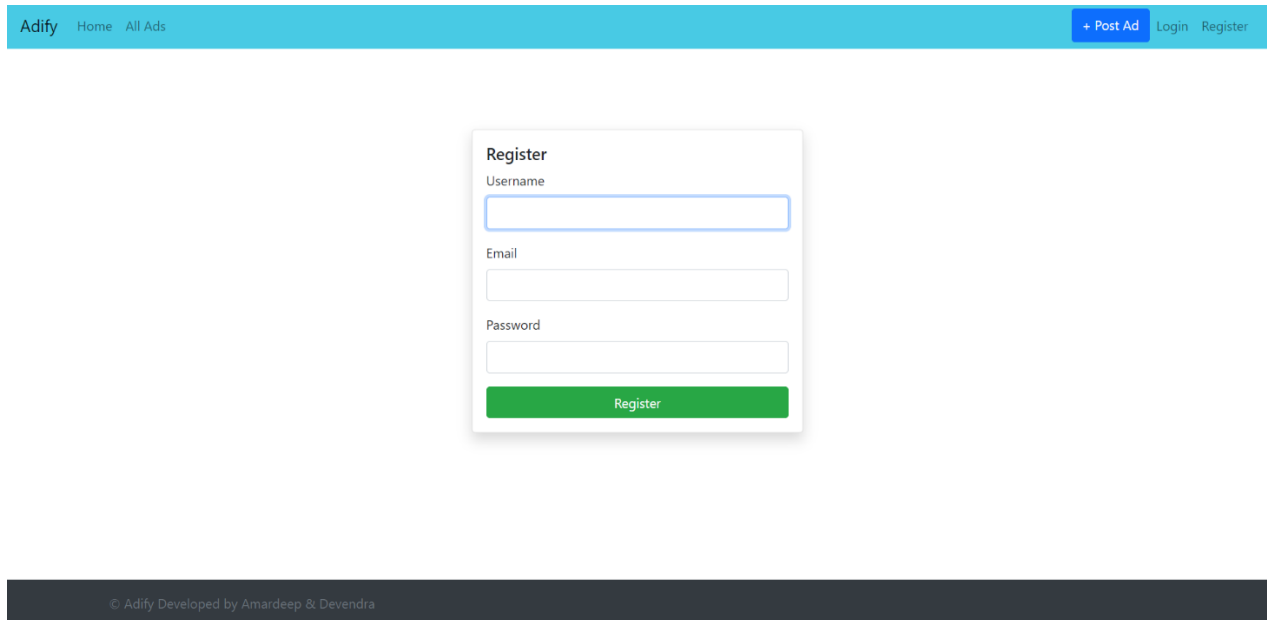
module.exports.login = (req, res) => {
    req.flash("success", "welcome back!");
    const redirectUrl = req.session.returnTo || "/ads";
    delete req.session.returnTo;
    res.redirect(redirectUrl);
};

module.exports.logout = (req, res) => {
    req.logout();
    // req.session.destroy();
    req.flash("success", "Goodbye!");
    res.redirect("/ads");
}
```


CHAPTER 5

RESULT

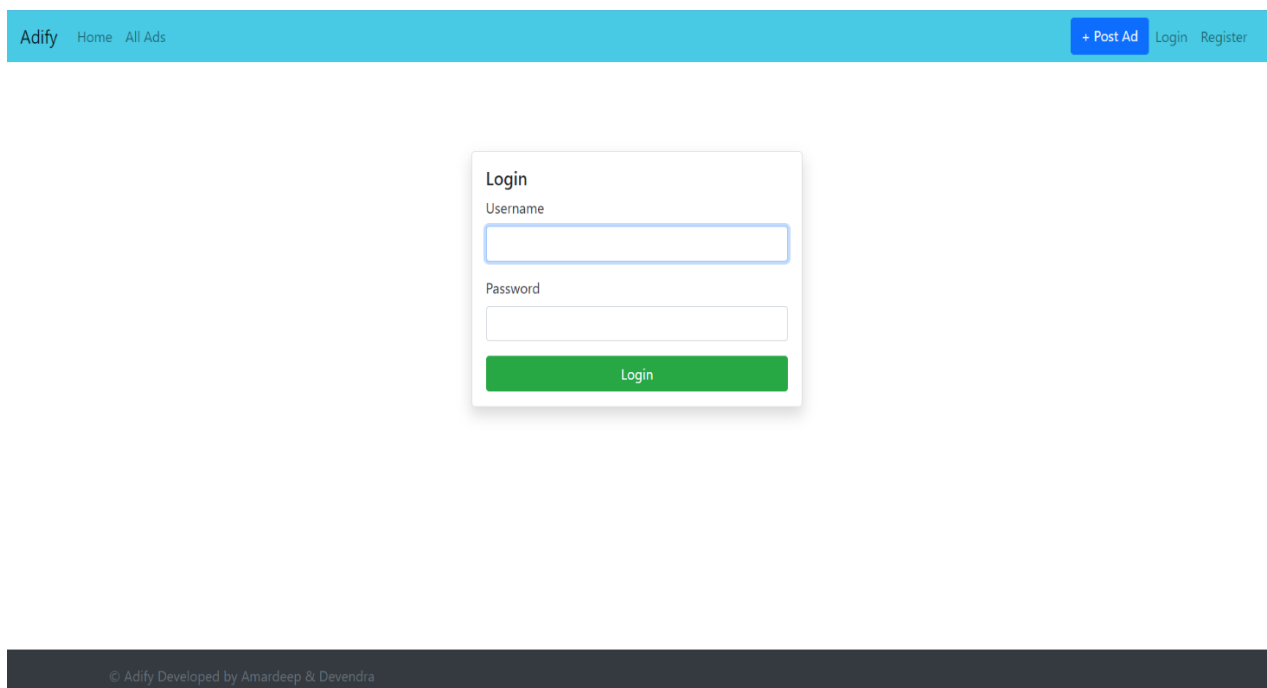
A new user can register by providing username,email and password



The screenshot shows the 'Register' page of the Adify application. At the top, there is a light blue navigation bar with the text 'Adify Home All Ads' on the left and a blue button '+ Post Ad' followed by 'Login Register' on the right. The main content area is white and features a 'Register' form. The form has three input fields: 'Username', 'Email', and 'Password'. Below these fields is a green button labeled 'Register'. At the bottom of the page, there is a dark grey footer bar with the text '© Adify Developed by Amardeep & Devendra'.

Figure 5.1 Register page

After registering, user can login by providing the credentials



The screenshot shows the 'Login' page of the Adify application. It has the same light blue navigation bar at the top as the register page. The main content area is white and features a 'Login' form. The form has two input fields: 'Username' and 'Password'. Below these fields is a green button labeled 'Login'. At the bottom of the page, there is a dark grey footer bar with the text '© Adify Developed by Amardeep & Devendra'.

Figure 5.2 Login page

Home page allows user with options such as all ads,login,register

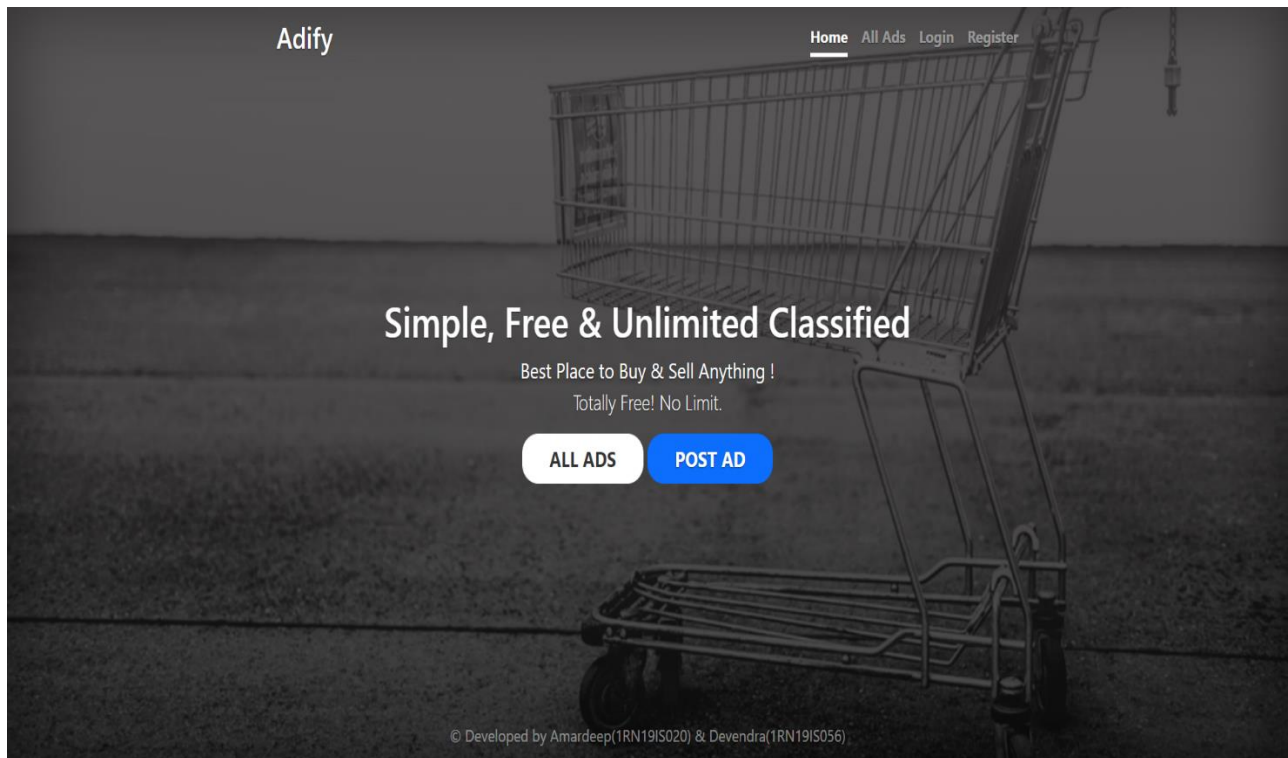


Figure 5.3 Home page

All ads page shows user with all the active advertisement posts

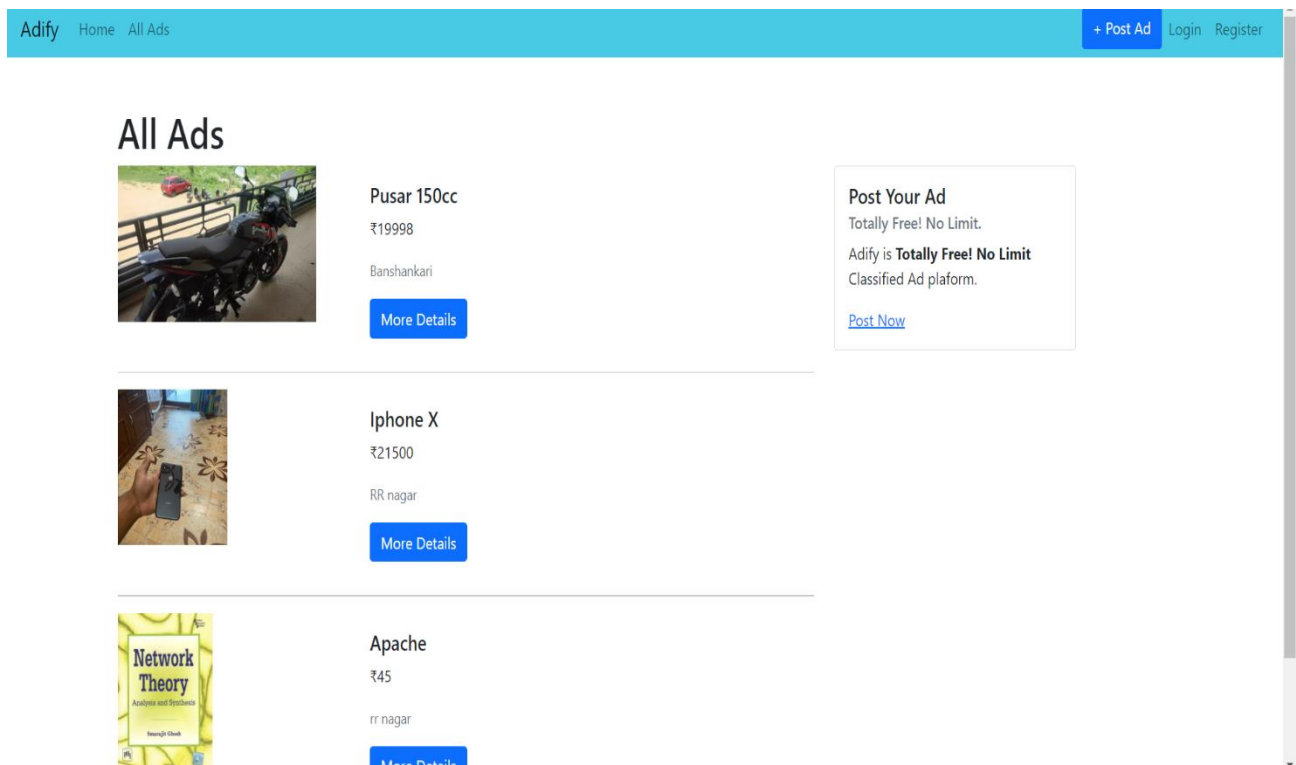


Figure 5.4 All ads page

Specific ad page by the owner shows edit and delete button

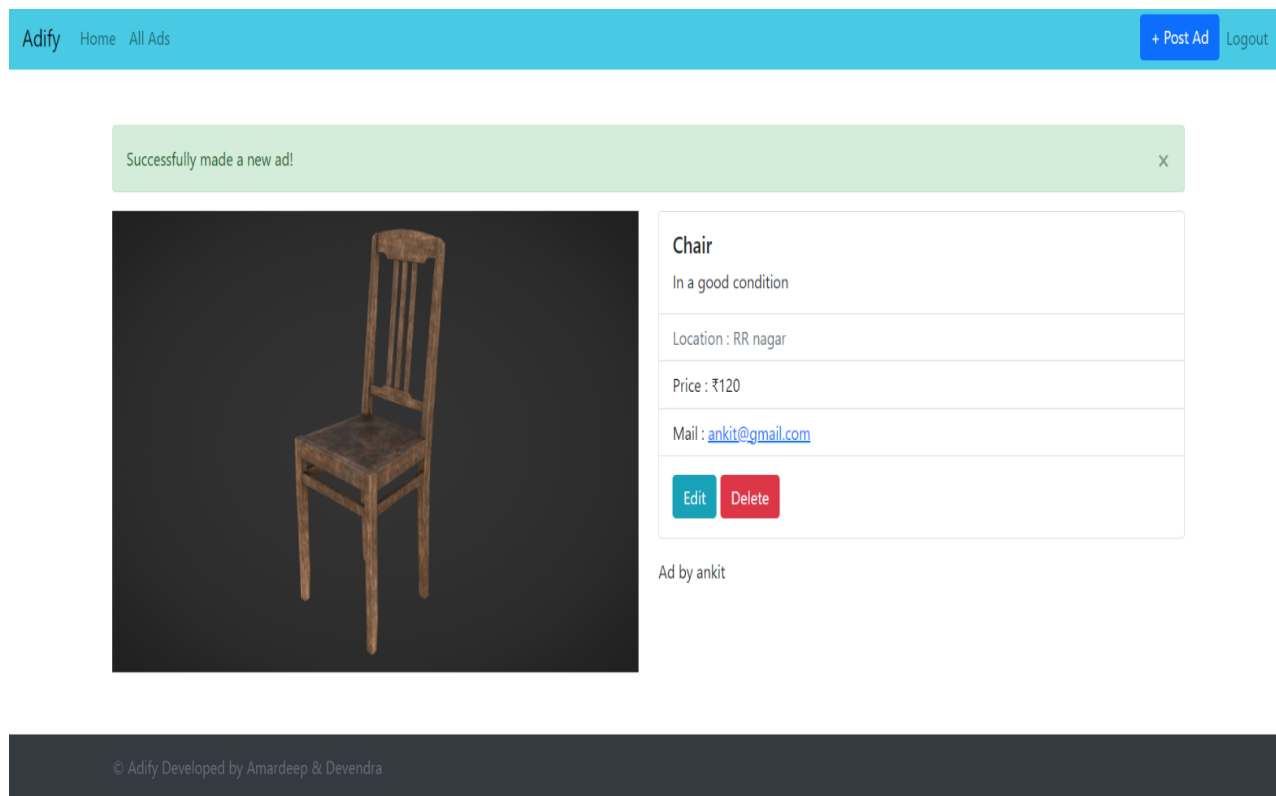


Figure 5.5 Specific ad page of same user

Specif ad page for user other than owner

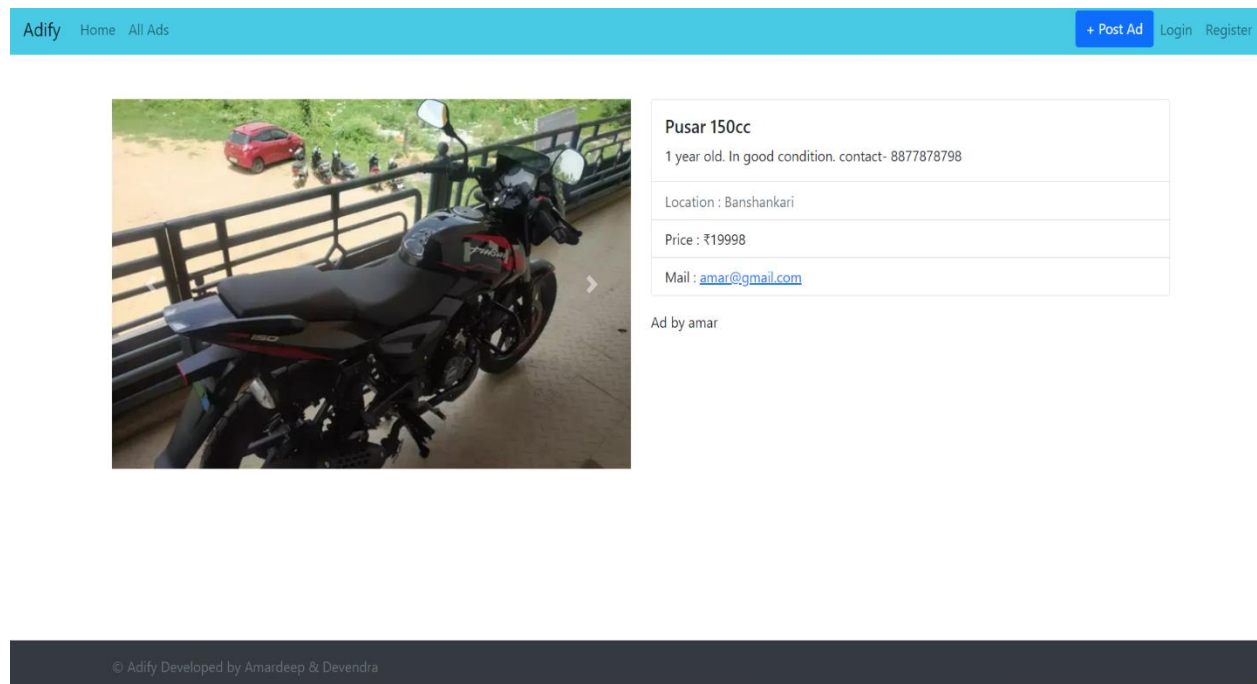


Figure 5.6 Specific ad page of different user

Post New Ad for registered user

Post New Ad

Title
Chair ✓
Looks good!

Location
RR nagar ✓
Looks good!

Ad Price
₹ 120 ✓

Description
In a good condition ✓
Looks good!

3.jpg Browse

Post Ad

Figure 5.7 Post new ad page

Edit page for the owner of the advertisement

Edit Campground

Title
Chair

Location
RR nagar

ad Price
₹ 120

Description
In a good condition

Add more image(s)... Browse

 ☐ Delete?

Update Ad

[Back To Ad](#)

Figure 5.8 Edit ad page

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion

This project was an attempt to make the structure and working of an Classified website simpler and user-friendly. This was an attempt to make it similar to the real world implementation. It helped us to understand the importance of planning, designing and implementation so far, we have learnt in our theory books. It helped us to understand MVC and EJS,NodeJS in detail. In this scenario, all the undertakings of the Online Marketplace was achieved in a constructive manner. Given the right guidance and support its applications and availability can be enhanced.

Future Enhancements

- Making the UI more interactive.
- Adding new features such as review system, where the buyer can rate the seller.
- Making the project more real time based by adding in app chat and transanction system.
- Hosting it on an online web server.

Chapter 7

REFERENCES

- [1] Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, McGRAW HILL, 3rd Edition.
- [2] Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems, Pearson, 7th Edition.
- [3] www.stackoverflow.com
- [4] www.mysql.com/doc
- [5] <https://docs.microsoft.com>
- [6] <https://www.geeksforgeeks.org>
- [7] <https://www.rapidapi.org>